S.C.E. HIGHER ANSWERS – 1996

PAPER I

1. $3x + 2y = 17$
2. $a = 2\frac{1}{2}$
3. $\left(\frac{5x}{6}, 3\right)$
4. $2x + y = 10$
5. 9
6. Verify that $\vec{AB} = \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$, $\vec{BC} = \begin{pmatrix} 6 \\ 3 \\ -3 \end{pmatrix}$

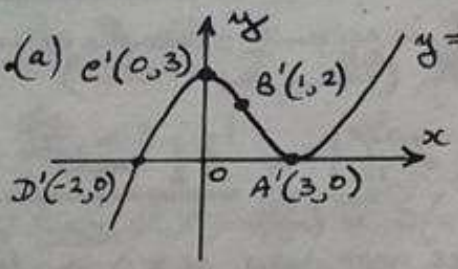
   $\therefore \vec{AB} = \frac{2}{3}\vec{BC}$

   $\therefore$ AB is parallel to BC

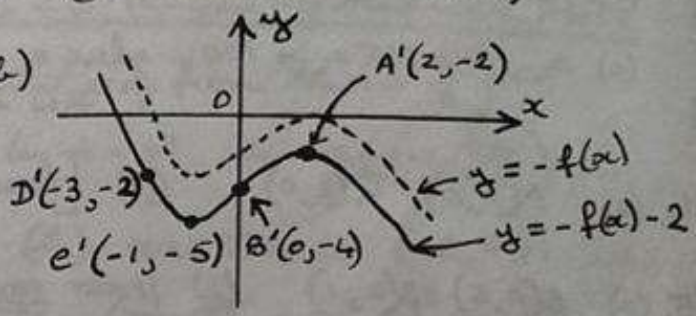
   $\therefore$ A, B, C are collinear [$\because$ B is a common point]

   B divides AC in the ratio 2 : 3 (i.e. AB : BC = 2 : 3)
7. $x(x-1)(x^2+x+1)$

8. (a)



(b)



9. $f'(4) = \frac{5}{16}$
10. $0°, 120°, 180°, 240°$ [NB NOT 360° since $0° \le x < 360°$]
11. (a) Limit exists because 0·3 lies between −1 and 1 (b) $7\frac{1}{7}$
12. $P\left(\frac{\pi}{6}, \frac{1}{2}\right)$ $Q\left(\frac{\pi}{2}, \frac{1}{2}\right)$
13. $\frac{dy}{dx} = \frac{-\sin x}{2\sqrt{1+\cos x}}$
14. Take 2 lines $\left\{\begin{matrix} 2x+3y=4 \\ 3x-y=17 \end{matrix}\right\}$. Verify that they intersect at (5, −2). Then verify that (5, −2) does NOT lie on the 3rd line $x - 3y - 10 = 0$

    $\therefore$ 3 LINES ARE NOT CONCURRENT.
15. [Verify that BD = 5 and AD = $2\sqrt{6}$; then use the expansion of $\cos(x+y)$.]
16. f is increasing when $x < -2$ AND $x > 3$ [TABLE OF VALUES REQUIRED]
17. $4(x+1)^2 - 9$
18. $\sin x = \frac{4\sqrt{5}}{9}$
19. (a) $k = 0.019$ (b) [FALLS TO 56·56°C IN NEXT 15 MINS]

    $\therefore$ IT FALLS 18·44°C IN NEXT 15 MINS
20. $(x-3)^2 + (y-4)^2 = 25$.

PAPER II

1. (a) S.P.'s are (0, 3) AND (3, −24) (b) (0, 3) is a point of inflection
   (3, −24) is a MINIMUM S.P.
2. (a) Verify that $m_{AB} m_{BC} = -1$ WITH A CLEAR CONCLUSION $\left(\begin{matrix} m_{AB} = 2 \\ m_{BC} = -\frac{1}{2} \end{matrix}\right)$

   (b)(i) AD: $x - 3y = 6$

   BE: $4x + 3y = -1$

   (ii) $M\left(1, -\frac{5}{3}\right)$

4. (a) $Q(2,2,9)$ $R(21,3,12)$ (b) $Q\hat{P}R = 83{\cdot}4^\circ$

(a) (i) $g[f(x)] = 4x^2+4x+1+k$ (ii) $f[g(x)] = 2x^2+2k+1$

(b) (i) [ANSWER IN QUESTION] (ii) [$\Delta = 64$] $\therefore$ Roots are REAL AND DISTINCT

(iii) $k = -2$

5. [area I $= \frac{1}{2} - \frac{\sqrt{3}}{4}$; area II $= \frac{1}{2}$] $\therefore$ TOTAL AREA $= 1 - \frac{\sqrt{3}}{4}$ units^2

6. (a) Line l : $y = 2x$ Parabola k : $y = 2x^2$ Circle c : $x^2+y^2 = 5$

(b) l' : $y = 2x-4$ k' : $y = -2x^2$ c' : $(x-2)^2+y^2 = 5$

(c) $(-2,-8)$ AND $(1,-2)$

7. (a) $\sqrt{13}\cos(x-56{\cdot}3^\circ)$ (b) $138{\cdot}3^\circ$, $334{\cdot}3^\circ$

8. (a) $5x+y = -3$; $B(-1,2)$

(b) $\frac{4}{3}$ units^2

9. (a) $Y = 3X + 0{\cdot}7$

(b) [FIRST REPLACE Y BY $\log_e y$ AND X BY $\log_e x$ TO OBTAIN :- $y = 2{\cdot}014x^3$.] $\therefore$ $\{a = 2{\cdot}014, b = 3\}$

10. (a) By symmetry, result needs only to be proved for one point

at $(3,7)$: PROVE THAT TANGENT TO PARABOLA HAS GRADIENT $\frac{2}{3}$

AND PROVE THAT TANGENT TO CIRCLE HAS GRADIENT $-\frac{3}{2}$.

Then conclude that since product of gradients of tangents is -1, the tangents are perpendicular. $\therefore$ CURVES ARE ORTHOGONAL.

(b) (i) $2y = 3x+23$

(ii) $x^2+(y-11\frac{1}{2})^2 = 29\frac{1}{4}$

11. (a) (i) $l = 5-x-\frac{1}{2}\pi x$

(ii) [START WITH $L = 2$(area of rectangle) $+ 1$(area of $\frac{1}{2}$ circle)]

(b) $x = \frac{20}{8+3\pi}$ $(\doteqdot 1{\cdot}1478)$

$h = \frac{20+5\pi}{8+3\pi}$ $(\doteqdot 2{\cdot}0493)$

NB : TABLE OF VALUES IS REQUIRED